OPC UA

Industrial Control Systems · SCADA Hub

COURSE DESCRIPTION

OPC UA is the successor to the classic OPC specifications and the standard communication backbone for Industry 4.0 and IIoT architectures. This course covers the client/server and pub/sub transport models, the node-based address space, the full service set, session and SecureChannel mechanics, and the certificate-based security model. Integration patterns using open-source stacks and commercial platforms are examined alongside Wireshark-based diagnostic techniques.

PREREQUISITES

Basic TCP/IP networking concepts. Completion of the Modbus course is recommended but not required.

COURSE TOPICS

1. **OPC UA Architecture:** Client/server and pub/sub transport models, the transition from DCOM, and the platform-neutral unified namespace.
2. **Information Model:** Address space, node classes, NodeIDs, type definitions, and the self-describing browsing model.
3. **Services:** Read, Write, Browse, Subscribe, and CreateMonitoredItems — the full service set and which matter in practice.
4. **Sessions and Channels:** SecureChannel negotiation, session establishment, token lifetime, and reconnection behavior on channel loss.
5. **Security:** Security Mode None vs. Sign vs. SignAndEncrypt; certificate exchange; PKI structure; consequences of disabled security.
6. **Subscriptions:** Publishing interval, MonitoredItem sampling, deadband filters, and the configuration chain controlling network traffic volume.
7. **Data Types:** Scalars, structures, enumerations, arrays, and the StatusCode quality system for representing sensor validity.
8. **Discovery:** Local and Global Discovery Servers, endpoint enumeration, certificate trust, and open-discovery security implications.
9. **Implementation:** SDK selection, Ignition as OPC UA server, common integration failures, and Wireshark-based diagnostic workflow.
10. **Protocol Lab:** Full session capture — TCP connection, SecureChannel, browse, subscription, disconnect, and session token renewal.

LEARNING OUTCOMES

- Browse an OPC UA server's address space and interpret node classes and references.
- Configure Security Mode Sign and SignAndEncrypt with certificate exchange on a production server.
- Build and tune a subscription with appropriate publishing intervals and deadband filters.
- Verify security mode and session token behavior from a Wireshark packet capture.
- Select and configure an OPC UA SDK or integration platform for a given deployment scenario.

REQUIRED SOFTWARE

- **open62541** (free, cross-platform) — open-source C OPC UA stack for server and client implementation exercises.
- **UA Expert** (free registration, Windows/Linux/macOS) — desktop client for address space browsing and subscription testing.